

Gamelan Semaradana Instrument Ranges

2 Jegogan	I	O	E	eu	U	A	ai
(7-key)	1	2	3	4	5	6	7

2 Calung	I	O	E	eu	U	A	ai
(7-key)	1	2	3	4	5	6	7

2 Penyacah	I	O	E	eu	U	A	ai	(1)
(7-key)	1	2	3	4	5	6	7	(1)

1 Ugal	O	E	U	A	I	O	E	eu	U	A	ai	I
(12-key)	2	3	5	6	1	2	3	4	5	6	7	1

4 Pemadé	O	E	U	A	I	O	E	eu	U	A	ai	I
(12-key)	2	3	5	6	1	2	3	4	5	6	7	1

4 Kantilan	O	E	U	A	I	O	E	eu	U	A	ai	I
(12-key)	2	3	5	6	1	2	3	4	5	6	7	1

1 Reyong (17-gong)	E	EU	U	A	AI	I	O	E	eu	U	A	ai	I	O	E	eu	U
(4 parts, divided as shown: vertical lines show conventional boundaries; horizontal lines show common extensions)	3	4	5	6	7	1	2	3	4	5	6	7	1	2	3	4	5
	<i>penyorog</i> (player 1)-----							----- <i>ponggang</i> (player 3)----									
								----- <i>pengenter</i> (player 2)-----							----- <i>pemetit</i> (player 4)		

Trompong	A	AI	I	O	E	EU	U	a	ai	I	O	E	eu	U
(14-gong)	6	7	1	2	3	4	5	6	7	1	2	3	4	5

Other instruments: 2 gong, 1 kempur, 1 kemong (hanging gongs); 1 kempli (seated gong); 1 cengceng (cymbals); 2 kendang (drum); 1-4 suling (bamboo flute).

Note that the jegogan, calung and penyacah have single-octave ranges, and together span three octaves. The ugal, pemadé and kantilan have two-octave ranges with the full seven-tone scale in the upper half only. Their ranges overlap to create a four octave span.

About this gamelan, its instruments, and its tuning.

In Bali and Java the word gamelan refers to a collection of instruments constructed and tuned to function as an inseparable set. This gamelan, of a type called *Gamelan Semaradana*, uses a tuning system commonly known by the Javanese name *pelog*, and also by the Balinese term *saih pitu* (series of seven). The full scale is of seven tones, but this is rarely used in any single composition. Instead one of several five-tone *patutan* (modes) is extracted from it. In the chart I, O, E, U and A indicate ding, dong, deng, dung and dang, the solfege designations for *selisir*, the

most frequently used patutan. The remaining two tones, eu and ai, (deung and daing) generally do not appear in selisir contexts. The full range of the gamelan--from lowest jegogan tone to highest kantilan tone--is 29 tones: four full octaves plus the first note of a fifth octave.

The numbers 1-7 shown in the chart below the solfege provide a convenient way to refer to the full seven-tone scale and the patutan that may be extracted from it, with ding of selisir conventionally counted as 1 because it is the leftmost (lowest) tone of the jegogan. However, unlike modern Javanese practice, the numbers are not used as the basis of a cipher notation system. (Balinese do have notation, based on Balinese script, but it is used mainly in manuscripts and occasionally by composers as a memory aid, but never by teachers or performers.) Other common patutan derivable from pelog include *tembung*, which uses the sequence 456 12, and *sunaren*, which uses 567 23. The same solfege system is applied to these and other modes too, with ding used to refer to the first tone of the patutan, dong to the second, etc., like in Western movable-do solfege. A total of seven modes is possible in pelog, each formed by taking one of the seven tones to be ding, taking the next two consecutive steps (dong and deng), skipping one (deung), taking two more (dung and dang), and skipping the last (daing). As with *tembung* and *sunaren*, above, this process may involve crossing over into a second octave.

As in many other Balinese gamelan, the issue of tuning in this ensemble is enriched by the Balinese fondness for tuning pairs of keyed instruments slightly apart from one another to create a rich acoustical beating audible whenever two corresponding tones are struck. This beating (*ombak-ombakan*) is held consistent throughout the range of the gamelan at a rate of approximately six or seven beats per second. To make this possible, octaves and other intervals need to be of slightly different sizes both between matching pairs of instruments and from octave to octave within the ensemble as a whole.

Gamelan Semaradana was innovated in the mid-1980s by the renowned gamelan maker I Wayan Beratha of Abian Kapas village in Bali, in collaboration with some younger musicians. This particular set of instruments was cast, forged carved, painted and assembled by Beratha and his helpers in Bali in 1989-90, and was one of the first of its kind built. Like most gamelan, the keys and gongs are made of a bronze alloy (*kerawang*) that is approximately three parts copper and nine parts tin. The wood is jackfruit (*nangka*). The panel carvings on the keyed instruments depict scenes from the *Ramayana* and *Mahabharata* on the fronts and backs respectively, which are the two best known and loved epic poems of classical Hindu culture throughout South and Southeast Asia. The hollow tubes that resonate the keys, hidden between the wood panels, are traditionally made from bamboo, but here are made from PVC plastic tubing as a safeguard against possible dryness and cracking in this

non-tropical climate. The gamelan was shipped to the U.S. in 1991 and played regularly from 1993-96 by Gamelan Jagat Anyar (Gamelan of the New World) in New Haven, Connecticut. Our Vancouver gamelan group goes by the name Gita Asmara (Love Song): www.gitaasmara.ca

The Semaradana combination of instruments and their ranges are closely modeled on the *Gamelan Gong Kebyar*, the preeminent type of Balinese ensemble since the 1930s. Kebyar gamelan are tuned exclusively to patutan selisir: in other words a Kebyar gamelan is exactly like a Gamelan Semaradana, with all instances of tones 4 and 7 deleted. Thus Kebyar jegogan, calung and penyacah have only five keys instead of seven; ugal, pemadé and kantilan have only ten instead of twelve, and so on. In Gamelan Semaradana the extra keys and extra gongs (for the reyong and trompong) were added to facilitate the playing of music from both the Kebyar repertoire and various pre-twentieth century Balinese repertoires (that use other patutan) on the same set of instruments--something that had never before been possible. In the older gamelan associated with these repertoires most instruments are limited to a one-octave range; that is why for the multi-octave keyed instruments ugal, pemadé, and kantilan the extra tones are present only in the upper register. The lower register is exactly as it would be in a Gong Kebyar. Beratha and his collaborators also intended to encourage new composition and experimentation combining the exuberant and fiery Kebyar style with the more refined and delicate older genres.

It is conventional among gamelan players to treat their instruments with appropriate deference. Balinese are culturally sensitive to their position in physical space as an indicator of prestige or respect. In the presence of esteemed people or things, one should always strive to situate oneself in a lower, more humble spot. Since the gamelan is considered spiritually charged and thus venerable, care is taken never to step over an instrument, but always to move around it, for one should never allow the soles of one's feet to face down towards it. I like to adopt this convention too, not because I am trying to be a pious Balinese, but because it creates an aura of awareness and respect for the instruments that is healthy and adds to rehearsal concentration and group unity. For the same reason, before performances offerings of fruit, flowers and incense are made to the gongs (the primary musical and spiritual focus), and the rest of the instruments and all the musicians and dancers are given libations of holy water.

For further general introduction to Balinese gamelan, see my book *Balinese Music* (Periplus 1998). For more detailed musicological studies, see Colin McPhee's *Music in Bali* (Yale Press 1966) and my *Gamelan Gong Kebyar: The Art of Twentieth Century Balinese Music* (University of Chicago, 2000). For the anecdotal experiences of a musician living in Bali during the 1930s, see McPhee's *A House in Bali* (Oxford Press 1979). I can provide further bibliographic references and suggest good recordings.

Gamelan Semaradana modal system and solfege:

Wayan Beratha, the composer and gamelan tuner, charts the most commonly accepted modes in relation to the seven tones of the scale as follows. Note that this terminology is not standardized throughout Bali and is indeed unknown in many places. They are shown compressed into a single octave:

1. <i>Selisir</i>	1	2	3	-	5	6	-
	I	O	E		U	A	
2. <i>Slendro Gdé</i>	-	2	3	4	-	6	7
		I	O	E		U	A
3. <i>Baro</i>	1	-	3	4	5	-	7
	A		I	O	E		U
4. <i>Tembung</i>	1	2	-	4	5	6	-
	U	A		I	O	E	
5. <i>Sunaren</i>	-	2	3	-	5	6	7
		U	A		I	O	E
6. <i>Pengenter alit</i>	1	-	3	4	-	6	7
	E		U	A		I	O
7. <i>Pengenter</i>	1	2	-	4	5	-	7
	O	E		U	A		I
8. <i>Lebeng</i>	1	2	3	4	5	6	7
	(N/A)						

(I = Ding, O = Dong, E = Deng, U = Dung and A = Dang)